

Homework — 2.1.4 Thinking Logically

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 2.1.4 lesson.

Part A — This week's topic (~14 marks)

1. A theme-park ride allows a rider on only if they are **at least 1.4 m tall**.

(a) Write the exact Boolean **condition** for the decision point. **[1]** (AO2)

(b) State the outcome of the TRUE branch and the FALSE branch. **[2]** (AO2)

2. A bank lets a customer withdraw from a cash machine only if the amount requested is **no more than** the balance **and** no more than the daily limit.

Write a suitable Boolean condition using a logical operator, and state what happens when the condition is **false**. **[2]** (AO2)

3. A water-tank controller turns a pump **ON** when the level falls below 20%, and **OFF** when it rises above 80%; otherwise it leaves the pump unchanged.

(a) Write the two Boolean conditions used at these decision points. **[2]** (AO2)

(b) State how many possible outcomes (including "leave unchanged") this logic has. **[1]** (AO2)

8. (1.4.1) Convert the denary number **45** into **8-bit binary**. [2] (AO2)